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Changes in point and diffuse source phosphorus inputs to the River Frome (Dorset, UK) from 1966 to 2006

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ABSTRACT

Changes in the relationship between soluble reactive phosphorus (SRP) concentration and river flow between 1966 and 2006 were assessed for the River Frome, UK using the recently developed Load Apportionment Model. The resulting source load estimates gave good agreement with known changes within the catchment. The model indicated an increase in point source contribution to the total river load from 46% to 62% between 1970 and 1985. This corresponded with the population increase within the catchment during that time. The predicted mean SRP load was highest between 1996 and 2000 (30 t y^{-1}), with 49% coming from point sources. Despite no lowering in population or major changes in agricultural practice, the model predicted a reduced load of 18.1 t y^{-1} for the period 2001 to 2005, due mainly to a decrease in point source inputs from 14.6 t y^{-1} to 6.1 t y^{-1} (equivalent to 34% of the total load). This prediction matches the major improvements in sewage treatment that occurred within the catchment in 2002. This study thus provides a major validation of the Load Apportionment Model. The model provides an effective and rapid method of determining past changes in phosphorus sources, based entirely on the P concentration – flow relationship: critically, it does not require any historical information on land use, fertiliser application rates, topography, soil types and sewage inputs. Further decreases in SRP concentration in the River Frome during the algal growing season would be best achieved by further reductions of STW inputs.

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1. Introduction

The quantification of phosphorus (P) loads entering nutrient-enriched rivers from point and diffuse (nonpoint) sources provides critical information to catchment managers, allowing them to select the most suitable nutrient mitigation options to reduce P concentration and eutrophication risk (Withers and Sharpley, 2008). Phosphorus source apportionment has traditionally been determined using either a source-oriented or load-oriented approach (EEA, 2005; White and Hammond, 2006). The source oriented approach estimates diffuse inputs using phosphorus export coefficients, and then sums these with the sewage input (derived either from direct measurement or a P export coefficient per person) to give an estimated total load (Bowes et al., 2005a; Hanrahan et al., 2001; Johnes, 1996). Such an approach requires detailed land use and agricultural census data, and knowledge of sewage treatment works (STW) phosphorus outputs. The load-oriented approach calculates the total load exported from the river, and subtracts the measured point inputs to give an

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estimate of diffuse inputs (Kronvang et al., 2005, 2007). This method of source apportionment requires accurate and frequent phosphorus concentration and flow data, and information on P discharges from all STWs upstream of the sampling point.

Long term phosphorus concentration data sets allow catchment managers to infer the effectiveness of previous P reduction measures, and play an important role in supporting future management decisions (Burt, 2003; Holmes, 2006). Applying source apportionment methodologies to such long term P concentration data sets greatly increases their usefulness. Thus, it allows the effect of past mitigation strategies to be directly quantified (Johnes et al., 2007), while providing the potential for predicting the outcome from the implementation of future P reduction measures. However, river water quality data sets spanning a number of decades are relatively rare in the UK (Heathwaite et al., 1996), and such nutrient data with an associated record of land use change, fertiliser application rates, animal stocking densities and sewage/industrial point inputs over the same time period are even rarer. Therefore these traditional methods of source apportionment are difficult to apply.

A new method of estimating the P loads derived from point and diffuse sources (the Load Apportionment Model) has been developed (Bowes et al., 2008) based on the fundamental difference in the characteristics of phosphorus inputs from these two source types. Point sources (from STWs and industry) provide a relatively constant input of P to the river throughout the year (Edwards and Withers, 2008). Therefore, rivers that are point-source dominated will have P concentrations that are highest at low river flow, and this concentration will decrease reciprocally with increasing flow rate, due to dilution. Conversely, diffuse P inputs (from manure and fertiliser inwash, soil erosion, septic tanks etc.) will be principally flow dependent, and therefore the phosphorus load increases with increasing flow in diffuse dominated rivers. The Load Apportionment Model determines the contribution of point and diffuse phosphorus inputs based entirely on the relationship between P concentration and river flow. It does not require any information on land use, fertiliser application rates, topography, soil types and sewage inputs, and so can potentially be applied to any long term dataset comprising of paired phosphorus concentrations and river flows. The model has been successfully applied to a range of catchments of varying sizes, geologies and land uses across England (Bowes et al., 2008), but has not yet been applied to investigate temporal changes in P source.

This paper provides an application of the Load Apportionment Model to a long-term soluble reactive phosphorus (SRP) concentration and flow record for the River Frome, Dorset, UK, a river that has been impacted by both agriculture and population drivers. There are four reasons for undertaking this study on the River Frome. Firstly, there is an excellent record of P concentration and flow information based on weekly sampling since 1966. Secondly, during the monitoring period there has been a significant increase in population within the catchment, and this will have resulted in an increase in P load during this time. Thirdly, phosphorus-stripping of the main sewage treatment works in the catchment was recently undertaken, thereby producing a known lowering of the P load. Fourthly, the River Frome has been part of a major hydrogeological and water quality study for the UK as part of the LOCAR programme (Wheater and Peach, 2004). Thus, this paper shows how phosphorus concentrations and

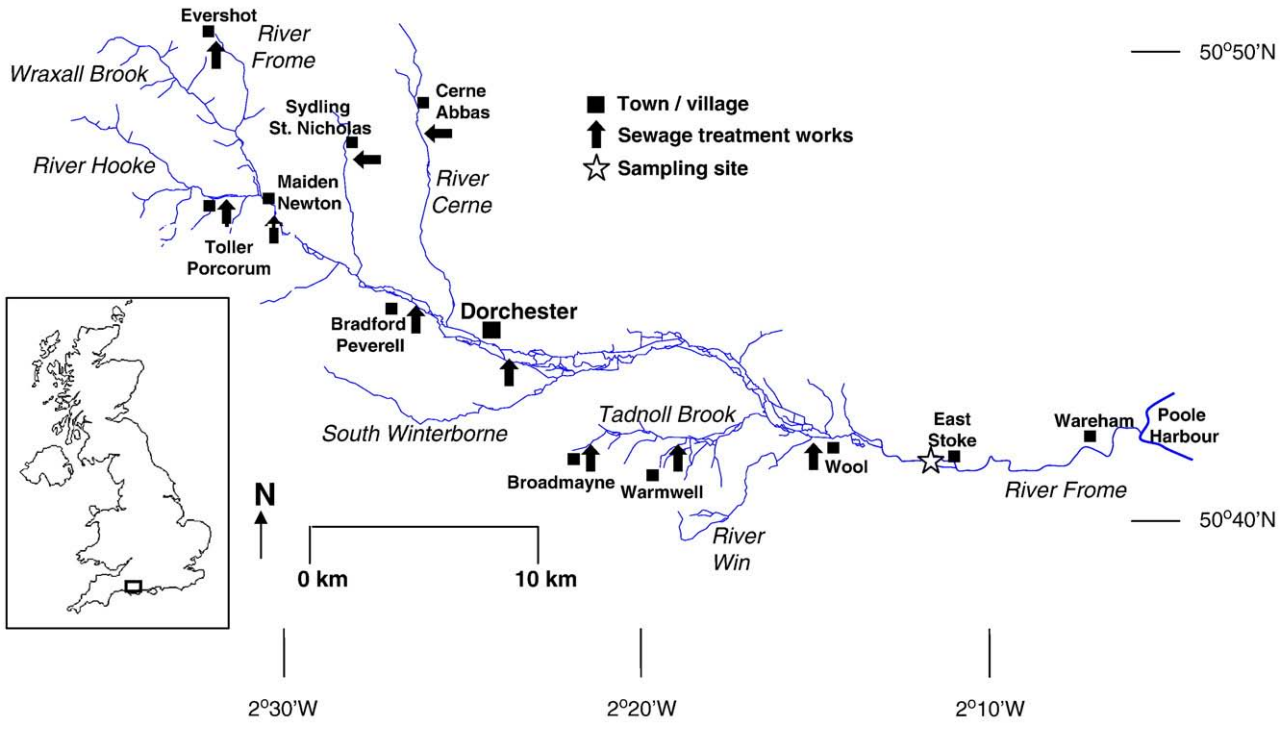


Fig. 1 – Map of the River Frome catchment.

Table 1 – Population estimates for sewage treatment works in the River Frome catchment, upstream of East Stoke, from 1994 to 2006

Sewage treatment works	Annual population estimate												
	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
Dorchester	21,388	23,012	22,301	22,699	23,238	25,862	23,495	23,484	24,200	25,841	25,627	21,472	26,623
Wool	3758	4315	5030	5030	5503	5469	6618	5476	5969	8434	8917	7900	11,912
Warmwell	3450	3450	3450	3450	3369	3365	3371	3273	6336	4776	4729	4706	0[§]
Bradford Peverell	2250	2250	2250	2251	2298	2306	2322	2213	2461	3495	3816	3820	3877
Maiden Newton	1469	1469	1469	1469	1500	1505	1515	1444	1596	1516	1663	1566	1614
Broadmayne	1425	1425	1425	1425	1455	1460	1470	1401	1484	1495	1494	1485	1473
Cerne Abbas	753	753	754	754	769	772	777	743	1863	1404	902	893	888
Sydling St. Nicholas	405	405	405	405	413	414	417	397	489	446	440	437	433
Toller Porcorum	240	240	240	240	245	271	273	260	405	332	346	288	283
Evershot	210	210	210	210	214	215	216	206	793	499	543	543	270
Total	35,348	37,529	37,534	37,933	39,004	41,639	40,474	38,897	45,595	48,238	48,477	43,109	47,374

Data provided by Wessex Water Services Ltd.

loads have changed with regards to population increase and subsequent remediation for a major community research catchment and it provides a critical evaluation of how the Load Apportionment Model has performed.

1.1. *Study area*

The River Frome catchment, Dorset, UK, covers an area of 414 km^2 (Casey and Newton, 1973), and extends from the village of Evershot (National Grid Reference ST 047576) on the Dorset – Somerset border, to Poole Harbour (Fig. 1). The underlying geology of the study area is Cretaceous Chalk, with areas of Cretaceous greensand in the River Hooke catchment. The land use within the catchment is primarily agricultural (mainly grassland and cereals) and the town of Dorchester is the only significant urban area in the catchment (Casey et al., 1993; Hanrahan et al., 2001). The study area and the locations of all STWs within the catchment are shown in Fig. 1. More detailed river and land use descriptions are given elsewhere (Casey et al., 1993; Hanrahan et al., 2001; O'Hare et al., 2007).

The water chemistry of the River Frome at East Stoke (National Grid Reference SY 867868) has been monitored at weekly interval since 1966 by the Freshwater Biological Association (1966 to 1988), Institute of Freshwater Ecology (1989 to 2001) and the Centre for Ecology and Hydrology (2002 to 2006). The daily average river flows are recorded at an Environment Agency gauging station at the East Stoke sampling site.

2. Materials and methods

2.1. *Analytical*

Water samples were collected from the River Frome at East Stoke at weekly intervals. A 100 ml aliquot of each sample was filtered through a 0.45μm filter and analysed for SRP concentration spectrophotometrically (Murphy and Riley, 1962). Sample filtering and SRP analysis were completed within 8 h of sampling, to minimise errors associated with sample instability (Haygarth et al., 1995; House and Warwick, 1998).

2.2. *Catchment information*

Background data on STW operations within the catchment since 1966 is limited. Known dates when phosphorus consents were reduced were provided by Wessex Water Services Ltd., to give an indication of the timing of improvements in water treatment and P removal. Population equivalents from all ten STWs were also available from 1994 to 2006 (Table 1). Information on population changes in the towns and villages within the catchment since 1961 were obtained from UK Census data (Table 2), and livestock numbers within the county of Dorset from 1999 to 2006 were obtained from the Department of Farming and Rural Affairs county level farming statistics (Defra, 2008).

2.3. *Load apportionment modelling*

A detailed description of the Load Apportionment Model and how it operates is given elsewhere (Bowes et al., 2008). In brief, phosphorus load (mg s^{-1}) from point, L_p, and diffuse, L_d,

Table 2 – Population of major towns and villages in the River Frome catchment from 1961 to 2001

Town/village	Census population				
	1961	1971	1981	1991	2001
Dorchester	12,263	13,740	14,050	15,100	16,180
Wool	3709	4130	5190	4330	4140
Broadmayne	598	870	860	1120	1180
Maiden Newton	680	690	760	910	950
Cerne Abbas	313	520	570	660	720
Frampton	367	360	390	400	470
Sydling St Nicholas	338	320	320	380	400
Bradford Peverill	301	340	350	320	330
Evershot	258	220	230	230	220
Total	18,569	20,970	22,490	23,220	24,370

Data from UK census information.

sources are modelled as power-law functions of the river volumetric flow rate, Q ($m^3 s^{-1}$):

$$L_p = A \cdot Q^B \quad \text{and} \quad L_d = C \cdot Q^D \tag{1}$$

The total load, L_t (mg s^{-1}) of P at the sampling point is therefore:

$$L_t = L_p + L_d = A \cdot Q^B + C \cdot Q^D \tag{2}$$

where *A*, *B*, *C* and *D* are load coefficients to be determined empirically.

Since the phosphorus concentration, C_p (mg m^{-3}) at a given sampling point is equal to the load divided by the volumetric flow rate, this can be expressed as:

$$C_p = A \cdot Q^{B-1} + C \cdot Q^{D-1} \tag{3}$$

The first term in Eq. (3) ($A \cdot Q^{B-1}$) is the P concentration originating from point sources, and the second term ($C \cdot Q^{D-1}$) is that part originating from diffuse sources. The four load coefficients in Eq. (3) are selected to provide the closest fit to the empirical P concentration and flow data, using the *Solver* function in Microsoft EXCEL©.

In many cases, a simpler model in which *B* was set to zero was also tested, as previous applications of this model to other rivers have shown that the inclusion of the *B* load coefficient often does not improve the model solution. Constraints were imposed on the model, to provide realistic solutions. Firstly, it was assumed that *B* must always be less than 1 (as the P concentration derived from point source inputs must decrease with increasing flow) and *D* values had to be greater than 1, as diffuse load inputs must increase with increasing river flow.

2.4. Load coefficient descriptions

The *A* load coefficient represents the load of phosphorus entering the river from all constant sources, which will principally include STWs, septic tanks connected directly to the river and, in some cases, industrial P inputs. This value will approximate to point source inputs. The *B* load coefficient describes processes such as the removal of phosphorus from the water column by adsorption to river bed sediments or biological uptake (Bowes and House, 2001; House, 2003). If *B*=0, the *A* load coefficient will have units of mg s^{-1}. The *C* load coefficient describes the load of P entering the river from diffuse (and therefore flow-dependent) sources. This will include diffuse P inputs from fertiliser and manure applications, animal faeces, soil erosion and septic tank soak-aways. The C load coefficient will also comprise of phosphorus that is stored in the river channel within the bed sediment or biota, which is remobilised during storm events. The model will apportion this phosphorus as coming from a diffuse source, but its original origin may be point source input during low flows, when sequestering by bed sediment, particular-bound P deposition and bioassimilation will be at their greatest. The rate of change of P load input with changing flow rate is described by the load coefficient *D*, and represents changes in the P concentration of storm-runoff/drain-flow, and the rate of within-channel P remobilisation, with increasing flow rate. Further descriptions of the load coefficients and how they respond are given elsewhere (Bowes et al., 2008).

2.5. Phosphorus load apportionment

The 40 year phosphorus concentration and flow data set was divided into eight five-year periods, as this was considered the smallest time interval that would still contain a representative range of river flows observed at the site. (Weekly sampling is likely to miss high flow storm events, resulting in bias in the data and an underestimate of P load) (Johnes, 2007; Jordan et al., 2007). Each five year data subset was modelled using Eq. (3). The river flow at which the estimated point and diffuse P inputs are equal (Q_e) were calculated using the equation:

$$Q_e = \left(\frac{A}{C}\right)^{(1/(D-B))} \tag{4}$$

When $Q < Q_e$, point sources will dominate the TP load, and diffuse inputs will dominate at river flows $> Q_e$.

Once the empirical data had been modelled for each 5 year period, the resulting *A*, *B*, *C* and *D* load coefficients were applied to the daily river flow data observed during the same five year monitoring period, thereby estimating the total load of phosphorus, T_p (mg yr^{-1}), flowing past the sampling point:

$$T_p = 86400. \sum_{i=1}^{i=365} A \cdot Q_i^B + C \cdot Q_i^D \tag{5}$$

Where Q_i is the mean daily volumetric flow rate ($m^3 s^{-1}$) and 86,400 is the number of seconds in one day. Eq. (5) again consists of both a point source ($A \cdot Q_i^B$) and a diffuse source ($C \cdot Q_i^D$) term. The results of the model fitting were used to determine the proportion of the SRP load that was contributed individually by constant (point) and flow-dependent (diffuse) phosphorus sources, throughout the 40 year monitoring period.

Much diffuse input of phosphorus will occur during episodic storm events, and a weekly sampling regime will therefore underestimate this diffuse contribution (Jarvie et al., 2002; Johnes, 2007; Jordan et al., 2007). Because the model (derived from weekly sampling) was applied to a daily flow data set, and the data subsets were of long enough duration to capture a range of high flow storm events, this bias in the data set is largely removed.

2.6. Model predictions and validation

The modelled solution for the period 1996 to 2000 was applied to the daily flow data observed between 2001 and 2006, to produce predicted SRP concentrations, assuming that point and diffuse inputs remained at 1996 to 2000 levels. In March 2002, phosphorus stripping was introduced at the Dorchester STW, resulting in a step reduction in point source input. The Load Apportionment Model was used to predict the effect of this step reduction in phosphorus point input by estimating the reduction in P load from the Dorchester STW, and re-calculating the *A* load coefficient (representing the total SRP load derived from all constant P inputs). The new model solution was then reapplied to the daily flow data from March 2002 to 2006, to predict the daily P concentrations in the River Frome following this STW improvement.

2.7. *Model limitations*

The model assumes that all point source inputs are continuous, and all diffuse inputs are flow dependent. While this is generally true, it will not be the case for all phosphorus sources. For example, phosphorus outputs from STWs will vary diurnally (Neal et al., 2000), due to changes in effluent loading into the STW through the day. Septic tank inputs are traditionally categorised as diffuse source (Carpenter et al., 1998), although in many areas of the UK, septic tanks are often directly connected to the river or land drainage system, and therefore operate as numerous small point source sewage inputs (Arnscheidt et al., 2007). The Load Apportionment Model will identify these inputs as point source, rather than the traditional diffuse classification.

3. Results and discussion

3.1. *Changes in SRP concentration*

The changes in SRP concentration throughout the 40 year monitoring period are shown in Fig. 2. There is a clear cyclical annual pattern, with the lowest SRP concentrations usually recorded during April and May, and the highest concentrations occurring between July and October. These peaks in bioavailable P concentrations during the ecologically sensitive plant growing period would be expected to have significant implications for ecological status and eutrophication risk (Bowes et al., 2007; Jarvie et al., 2006).

Throughout the monitoring period, the River Frome has had a relatively high SRP concentration, exceeding the Environment Agency's 60 $\mu g\ l^{-1}$ annual average target concentration for chalk rivers (Mainstone and Parr, 2002). The annual average SRP concentrations steadily increased from 90 to 158 $\mu g\ l^{-1}$ between 1966 and 1973, with an associated increase in maximum concentrations of 139 to 237 $\mu g\ l^{-1}$ respectively (Fig. 2b). From the mid 1970s until 2000, the average SRP concentration remained at *ca.* 140 $\mu g\ l^{-1}$, with peaks (up to 190 $\mu g\ l^{-1}$) occurring in the low flow years of 1976, 1984, 1989 and 1992. In 2002, there was a step decrease in both average and maximum observed SRP concentrations, with an average SRP concentration of 88 $\mu g\ l^{-1}$ between 2002 and 2005. This decrease corresponds with the introduction of phosphorus stripping at the Dorchester STW in March 2002.

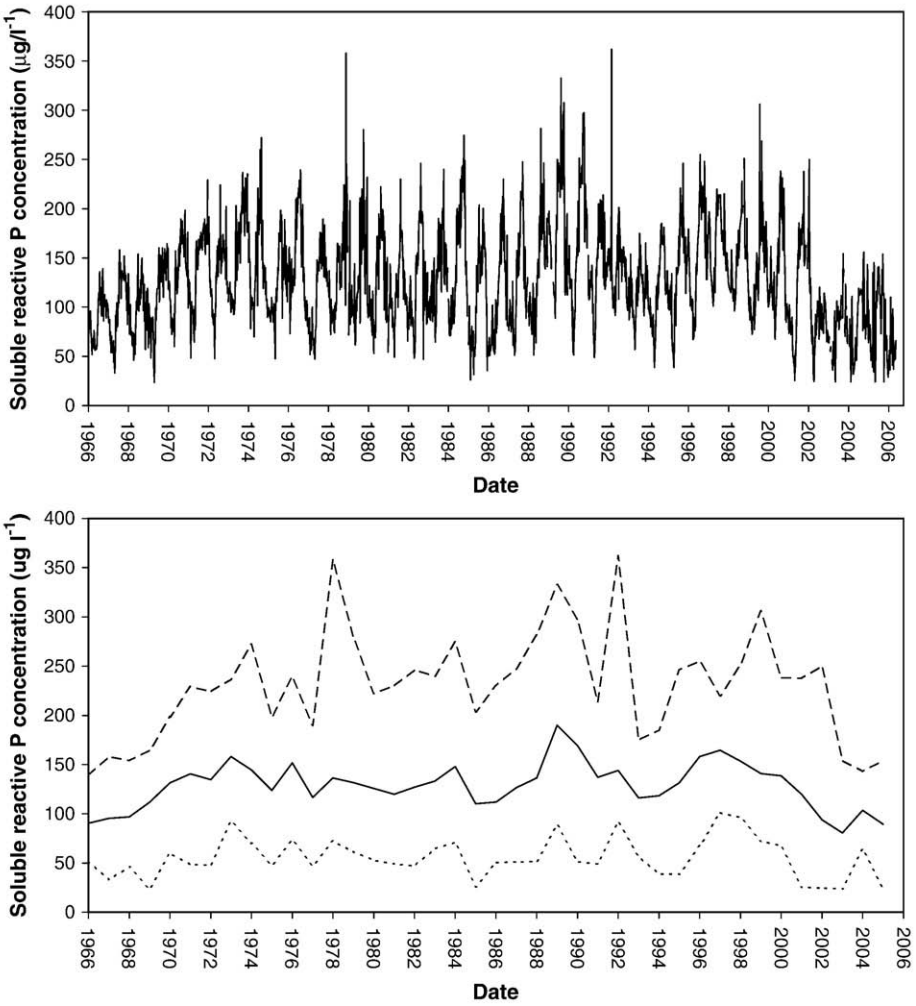


Fig. 2 – Soluble reactive phosphorus concentrations in the River Frome at East Stoke, 1966 to 2006. (a) Weekly data. (b) Annual mean (solid line), maximum (dashed line) and minimum (dotted line) soluble reactive phosphorus concentrations.

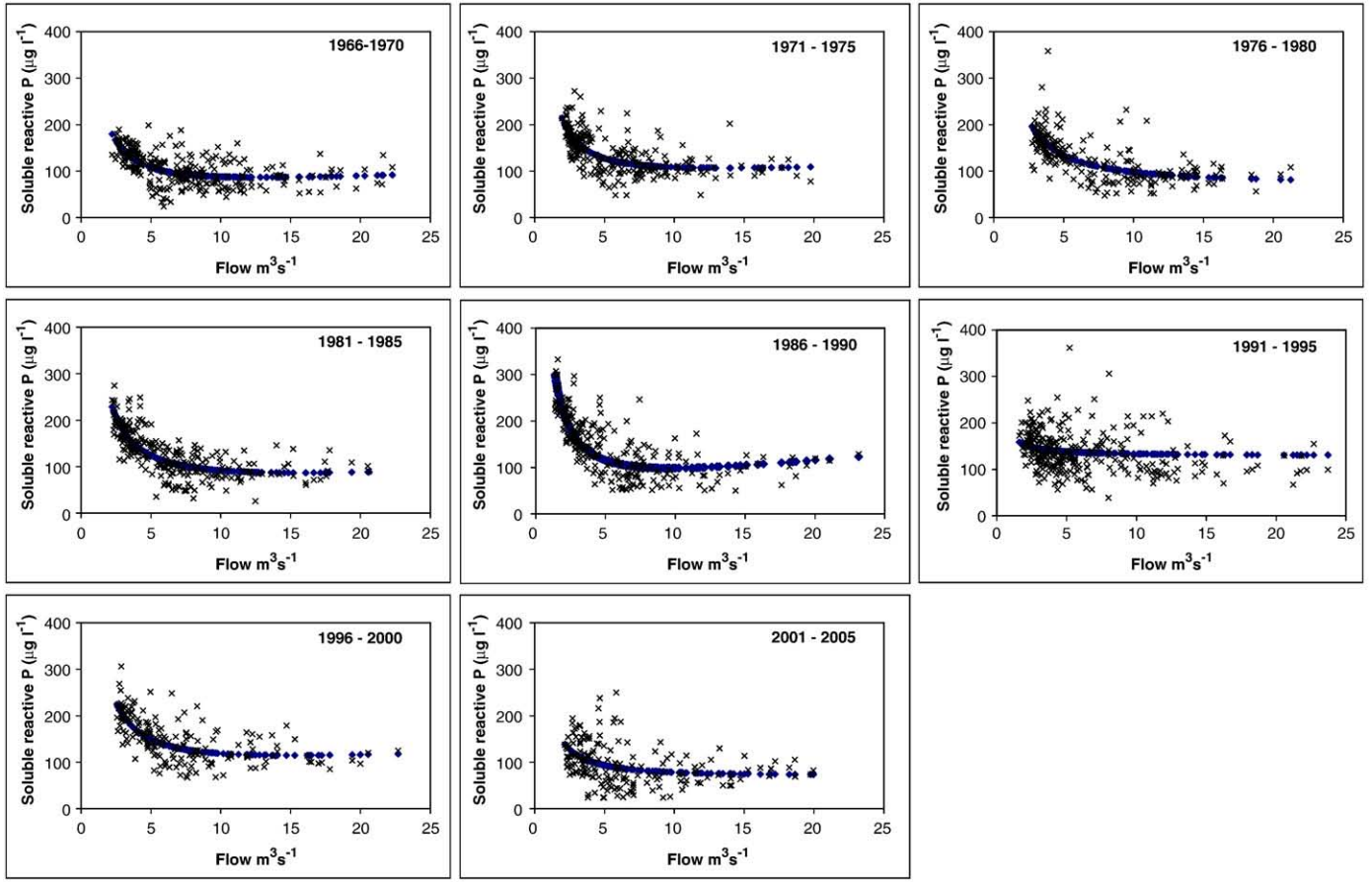


Fig. 3 – Relationship between soluble reactive phosphorus (SRP) concentration and flow for the River Frome catchment at East Stoke. *x* = measured river SRP concentration data. ♦ = combined SRP concentrations from point and diffuse sources, using Load Apportionment Modelling.

Table 3 – Results of Load Apportionment Modelling of the five-year data subsets

Date	Average river flow ($m^3 s^{-1}$)	Load coefficients				Proportion of samples point source dominated (%)	Point source load ($t y^{-1}$)	Diffuse source load ($t y^{-1}$)	Estimated total load ($t y^{-1}$)	Proportion of load from point source (%)
		A	B	C	D					
1966–1970	7.6	337	0	20.2	1.43	54	10.6	12.5	23.1	46
1971–1975	5.7	330	0	38.2	1.29	56	10.4	12.3	22.7	46
1976–1980	6.9	394	0	47.7	1.09	54	12.4	12.4	24.8	50
1981–1985	6.8	462	0	14.0	1.51	83	14.6	8.9	23.5	62
1986–1990	5.7	396	0	11.8	1.70	80	12.5	9.2	21.7	58
1991–1995	6.1	No solution	–	–	–	–	–	–	–	–
1996–2000	7.1	464	0	31.5	1.36	85	14.6	15.4	30.0	49
2001–2005	6.6	193	0	45.6	1.11	44	6.1	12.0	18.1	34

The average river flows for each 5 year period is based on weekly flow data at the time of sampling.

3.2. *Model outputs*

The Load Apportionment Model produced realistic fits to the observed data for seven of the eight of the five year data subsets (Fig. 3). The resulting load coefficient values are given in Table 3, and the estimated point and diffuse source contributions to the total SRP load (from Eq. (5)) are given in Fig. 4. The optimal model solutions were obtained using only three of the four load parameters; *A*, *C* and *D*. The *B* parameter (representing within-channel retention and release processes acting on the constant SRP inputs, and variations in load inputs from STWs) did not improve the model solution, indicating that point inputs were relatively constant through the year, and that internal phosphorus dynamics were relatively unimportant in determining SRP concentrations, in comparison to the input loadings from point and diffuse sources in the catchment.

The model was unable to find a satisfactory solution for the period 1991 to 1995, as there was not a clear relationship between P concentration and flow (Fig. 3). Therefore, this model solution was rejected. This five-year data subset had the largest range of mean annual flows, ranging from the lowest flow in the 40 year data set (3.8 $m^3 s^{-1}$ in 1992) to one of the highest (8 $m^3 s^{-1}$ in 1994), which may account for some of the scatter in the SRP concentration – flow relationship. There is an underlying relationship between SRP concentration and flow during this period that was similar to the other five-year data subsets, with high concentrations at the lowest flows, and a subsequent dilution with increasing flow. However, the highest SRP concentrations occurred sporadically during medium flow conditions (between 5 and 7 $m^3 s^{-1}$), which does not fit with our assumption that SRP input to the river is either from a constant or a flow dependent source. This may indicate that the phosphorus input is diffuse dominated, and that this source becomes depleted at flows above 8 $m^3 s^{-1}$, thereby resulting in reduced P concentrations at the higher river discharge. This explanation is, however, unlikely, as the SRP concentrations at higher flows (>12 $m^3 s^{-1}$) remain relatively constant, indicating increasing P load with increasing river discharge and therefore there cannot be any catchment phosphorus depletion. It is more likely that there is a significant input of phosphorus from random pollution

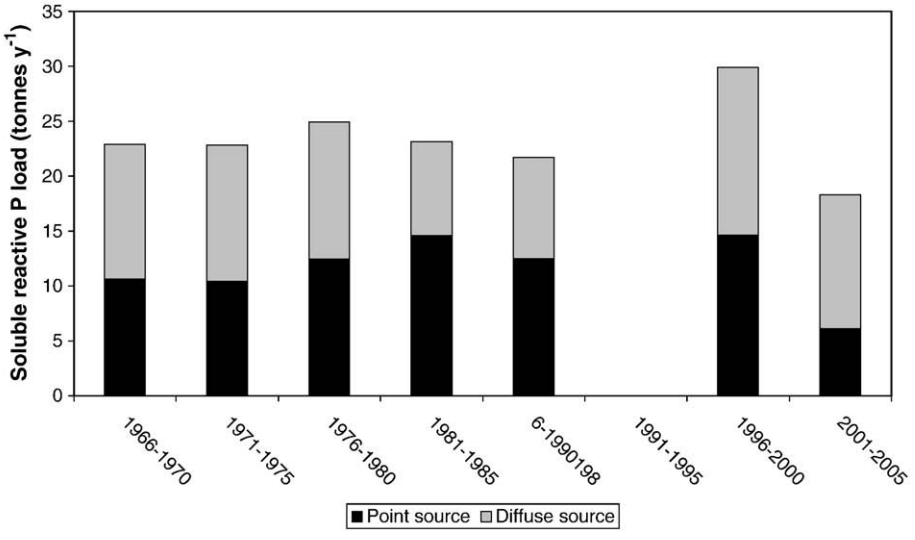


Fig. 4 – Point and diffuse source contributions to the total SRP load between 1965 and 2005.

incidents that are neither constant nor related to flow, resulting in a poor relationship between SRP concentration and flow rate. Similar observations and conclusions have been made by Jordan et al. (2007) for the Lough Neagh basin, Northern Ireland. In a previous study, the Load Apportionment Model was applied to a total phosphorus concentration data set for the River Frome at East Stoke in 1999 to 2000, and was also unable to fit the empirical data, due to even more pronounced non-flow-related phosphorus inputs (Bowes et al., 2008). This strongly indicates that there is a source of primarily particulate-bound phosphorus input to the lower River Frome that is neither constant nor flow related.

3.3. *Changes in SRP source*

SRP concentrations were greatest during periods of low flow for all eight five-year subsets of data (Fig. 3). This explains why there is a clear annual cycle in SRP concentration throughout the 40 year dataset, with the highest P concentrations always occurring during the July to October period, when flows in the River Frome are at their lowest. This type of concentration – flow relationship is indicative of a significant input of phosphorus from point sources (Cooper et al., 2002; Jarvie et al., 2006; Jordan et al., 2007; Wood et al., 2005).

The A load parameter values derived from the Load Apportionment Model (representing the estimated input of SRP from constant sources in mg s^{-1}) for each five year data subset indicate that there has been a *ca.* 40% increase in point source inputs of SRP between 1970 and 1985 (Table 3) (Fig. 4). This is equivalent to a predicted increase in annual point source input of 4 t y^{-1}. This correlates with the increasing resident population in the catchment throughout the study period (Table 2). There are no records of STW population estimates or sewage effluent P concentration data available during this period in the River Frome catchment, but similar conclusions about increasing point inputs to the River Frome during this period have been made by Casey et al. (1993), based on STW effluent volumes. This increase in point source SRP input is most likely due to the increasing population within the catchment (Table 2), and potentially an even greater increase in population density due to growing tourism in the county. Another contributing factor could be improvements to the wastewater infrastructure, resulting in remote villages and settlements (that were previously served by septic tank systems) becoming connected to the sewerage network. This potential decrease in septic tank usage may also account for some of the 3 t y^{-1} reduction in the estimated diffuse input to the River Frome during the 1980s (Table 3). The quantity of phosphate fertiliser use across Britain began to decrease from 1983 (Defra, 2002), and the period 1986 – 1990 was relatively dry with low river flows (Table 3), which may also account for the reduction in diffuse SRP input to the river. As there were no detailed agricultural or land use statistics available for the River Frome catchment prior to the 1990s, this reduction in diffuse load can not be further investigated.

The Load Apportionment Model predicted that there was a 2.1 t y^{-1} decrease in point inputs of SRP in the period 1986 to 1990, despite a continuing increase in human population in the catchment. This corresponds with the improvements in the level of sewage treatment and reductions in phosphorus consents at five of the largest STWs (Dorchester, Wool, Maiden Newton, Broadmayne and Warmwell) between 1989 and 1990 (Ruth Barden, Wessex Water Services Ltd., pers. com). Between 1991 and 2000, the only STW improvements occurred at the two smallest STWs at Evershot and Toller Porcorum. During this same period, there was an increasing population in the catchment (Table 2), with the total population equivalent for all ten STWs in the catchment increasing from 35 348 in 1994 to 40 474 in 2000 (Table 1). Therefore, the model prediction of increasing SRP input from point source during the 1990s agrees with the available data for the catchment.

To verify these model outputs against some external empirical data, the modelled estimates of point source loads (derived from Eq. (5)) (Table 3) were divided by an export coefficient for P output from STWs of 0.38 kg ca^{-1} y^{-1} (for primary and secondary sewage treatment) (Johnes, 1996), to give estimates of total population equivalents (p.e.) of the catchment's STWs. These were then compared to the actual population equivalents available since 1994 (Table 1). The model predicted that the total p.e. of the catchment was 27 969 in the late 1960s, and increased to 38 536 in 1996 – 2000. The actual average p.e. for the period 1996 – 2000 was 39 317, which demonstrates the accuracy and potential usefulness of the Load Apportionment Modelling approach in estimating point source loading.

The modelled SRP load was highest for the 1996 – 2000 data subset, with an estimated export of 30 t y^{-1} (Fig. 4). The estimated point and diffuse inputs were also both at a maximum, at 14.6 and 15.4 t y^{-1} respectively (Table 3). This period had relatively high annual mean flows, varying from 5.9 m^3 s^{-1} to 8.7 m^3 s^{-1} in 1997 and 2000 respectively, which would be expected to increase the diffuse phosphorus input. In contrast, the model produced the lowest estimated total load (18.3 t y^{-1}) in 2001 to 2005, due mainly to a marked decrease in point inputs from 14.6 t y^{-1} to 6.1 t y^{-1}. This reduction in modelled point source input coincided with major improvements in sewage treatment within the catchment. Phosphorus stripping was introduced at Dorchester STW in February 2002. Further reductions in phosphorus consents at the Bradford Peverell, Warmwell and Sydling St. Nicholas STWs were introduced in 2004, 2002 and 2001 respectively, and the Warmwell STW was closed in 2005, with routing of its effluent through the improved Dorchester STW.

The model estimated that the proportion of the total SRP load derived from point sources (using Eq. (5)) increased from 46% in the period from 1966 to 1975, to 62% in the mid 1980s. Following STW improvements in 1989 to 1992, the proportion decreased to 49% in 1996 to 2000. This is a higher STW contribution than the 35% estimated by Hanrahan et al. (2001) for the River Frome in 1998, using a traditional export coefficient approach, although this was based on total phosphorus concentrations. In this study, the model has been applied to SRP concentrations, rather than total phosphorus (as total P concentration was not monitored on the River Frome until 1991). The SRP fraction provided the majority of the total phosphorus load in the River Frome during the period when TP data was available (1991 – 2003), with an average contribution of 65% (Bowes et al., 2005c; Hanrahan et al., 2001). It is important to note that modelling only the SRP fraction will affect the estimates of source apportionment, as STW inputs are predominantly in the SRP form and much diffuse source P input will initially enter the river in non-bioavailable particulate-bound form. SRP is usually assumed to

be equivalent to bioavailable P (Nurnberg and Peters, 1984), and is therefore the phosphorus fraction that will directly affect eutrophication risk.

3.4. *The use of the Load Apportionment Model as a predictive tool*

The Load Apportionment Model has produced realistic predictions of the changes in the relative contributions from point and diffuse SRP sources which match the limited sewage treatment information and population data available for the River Frome catchment over the course of the 40 year study period. However, to truly test the model and its usefulness, it must be used to retrospectively predict the impact of a known change in phosphorus source input. The introduction of phosphorus stripping at Dorchester STW in March 2002 allowed the model to be tested in this way.

The model solution for the 1996 to 2000 data subset was applied to the average daily river flows observed in the period 2001 to 2006, to give predicted SRP concentrations if the point and diffuse inputs had remained at 1996 to 2000 levels (Fig. 5). The model predicted that the SRP concentration would reach a maximum of *ca.* 250 $\mu g\ l^{-1}$ each year, during the September/October low flow periods. During winter high flows, the SRP concentration was predicted to be *ca.* 120 – 140 $\mu g\ l^{-1}$ with a minimum concentration of 115 $\mu g\ l^{-1}$. Previous nutrient limitation experiments on the River Frome at East Stoke have highlighted that SRP is in excess for periphytic algal growth at concentrations above 90 $\mu g\ l^{-1}$ (Bowes et al., 2007). Therefore there would be a danger of eutrophication and algal blooms during periods of high temperature and light levels under this scenario, as phosphorus would be in excess for algal growth throughout the 2001 to 2006 period.

The model was then used to predict the SRP concentrations following phosphorus stripping at Dorchester STW. The total SRP load entering the River Frome from all STWs in the catchment in 2001 (i.e. prior to the introduction of P stripping in February 2002) was estimated, by multiplying the total p.e. of all STWs by a phosphorus export coefficient of 0.38 kg $ca^{-1}\ y^{-1}$ for human sewage treated through primary and secondary sewage treatment processes (Johnes, 1996). This gave an estimated point source P input of 14.8 t y^{-1}, which was in close agreement with the estimate obtained from the Load Apportionment Model for 1996 to 2000 of 14.6 t y^{-1} (Table 3). Using the same export coefficient approach, the output from Dorchester STW in 2001 was estimated to be 8.9 t y^{-1}, (equivalent to 60% of the total point source P input to the river). For the period following the introduction of P stripping at Dorchester, it was assumed that there would typically be a *ca.* 80% reduction in phosphorus in the STW effluent (Bowes et al., 2005a; Jarvie et al., 2002), reducing the output from Dorchester STW to 1.8 t y^{-1} and the total output from all STWs would be reduced to 7.6 t y^{-1}. This is equivalent to a reduction in total STW input to the River Frome of 52%. This is again in good agreement with the estimated point source load derived from the Load Apportionment Model for 2001 to 2005 of 6.1 t y^{-1} (Table 3).

The Load Apportionment Model was rerun to determine the effect of this estimated 52% decrease in point source input. The value of the *A* load coefficient from the 1996 to 2000 was reduced by 52%, and then this model solution (with load coefficients of *A*=241; *B*=0; *C*=31.5; *D*=1.36) was reapplied to the daily average river flow data from March 2002 (the date when the sewage treatment improvements commenced) to May 2006. The estimated SRP concentrations resulting from the phosphorus stripping at Dorchester are given in Fig. 5. The model produced predictions of summer SRP concentrations that were in close agreement with those observed from the chemical monitoring. It correctly predicted that the maximum SRP concentrations would fall from 250 $\mu g\ l^{-1}$ (prior to P stripping) to 150 $\mu g\ l^{-1}$. The model estimated that the point source contribution to the total SRP load would be reduced from 49% in 1996 to 2000 to 38% after February 2002.

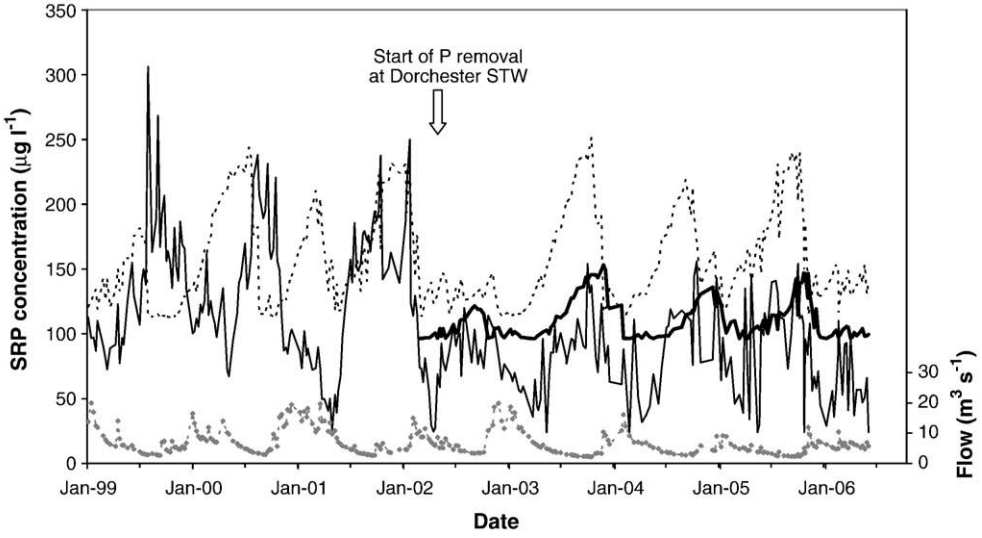


Fig. 5 – Predicted effect of phosphorus removal at Dorchester sewage treatment works (STW) on the soluble reactive phosphorus (SRP) concentration in the River Frome at East Stoke. Solid line = measured SRP concentrations. Bold solid line = predicted SRP concentration following P removal at Dorchester STW. Dashed line = predicted SRP concentrations if P removal had not taken place. ♦ = river flow.

The predictions of SRP concentration during the high flow periods were less precise than those observed during the summer, with the model consistently overestimating the P concentrations during winter and spring. This is because the model assumed that the diffuse input rate would remain the same as in the period 1996 – 2000. There was a step decrease in SRP concentration in early 2001 (Fig. 5), and this is most likely due to a reduction in flow-dependent diffuse inputs (as the model successfully fits the empirical data in the subsequent low flow period from mid 2001 to early 2002). This decrease coincides with the Foot and Mouth livestock disease outbreak in the UK in 2001, which resulted in a sudden and sustained change in land use in the county of Dorset, with a 9% decrease in dairy and beef cattle (from 91534 in 2001 to 83511 in 2002), and a 68% increase in sheep numbers (from 105598 in 2001 to 176598 in 2002) (Defra, 2008). This trend in land use change continued in the following years, with sheep numbers reaching 193455 and cattle numbers further decreasing to 77532 by 2006.

3.5. *The Load Apportionment Model as a catchment management tool*

The Load Apportionment Model has shown how previous mitigation options and changes in the River Frome catchment, such as sewage treatment improvements, population increases and changes in farming practises, have had an impact on the relative SRP contributions from point and diffuse sources. It has also been demonstrated within this study that the model can predict the SRP concentrations through the year following the introduction of a particular phosphorus mitigation option. Because the model produces high-resolution temporal data, it can provide a powerful tool in determining the most appropriate phosphorus mitigation strategies to apply to further improve water quality and minimise eutrophication risk.

Once the model has provided a realistic solution to the SRP concentration and flow relationship for a river, the flow at which point and diffuse source inputs to the river are equal (Q_e) can be calculated, using Eq. (4). This Q_e value provides a valuable aid for effective nutrient management of the catchment. For example, the model predicted that 1981 to 1985 was the period when the River Frome was most dominated by STW inputs within the 40 year data set, with point sources providing 63% of the total SRP load to the river (Table 3). However, in terms of controlling eutrophication, it is the nutrient concentrations (rather than loads) during the plant and algae growing seasons that control nuisance epiphytic algal growth, and so are critical for catchment managers to consider (Hilton et al., 2006). The relationship between SRP concentration and flow for the period 1981 to 1985 is shown in Fig. 6a. The data points were divided into two equal

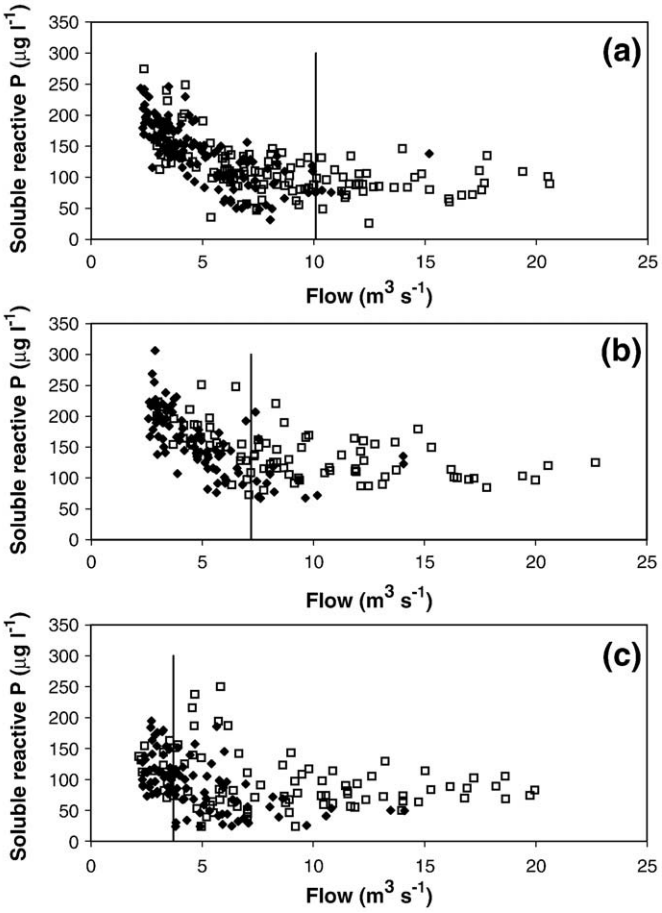


Fig. 6 – Seasonality of soluble reactive phosphorus inputs. (a) 1981 to 1985. (b) 1996 to 2000. (c) 2001 to 2005. ♦ = samples taken during the main plant and algae growing season (1st April and 30th September) □ = samples taken between 1st October and 31st March. Solid vertical line = Q_e (the river flow at which the estimated point and diffuse inputs are equal).

groups based on the time of sampling; 1st April to 30th September (a six month period roughly equating to the peak plant and algae growing season in temperate climates in the northern hemisphere) and October 1st and March 31st. For the 1981 to 1985 data subset, the Q_e value was calculated to be 10.1 $m^3 s^{-1}$ (Table 3). All the highest SRP concentrations occurred at low river flows, with the majority (83%; Table 3) of the data points and 97% the summer growing season data points occurring at flows lower than the Q_e value (Fig. 6a), when point source inputs are dominating. Therefore it is clear that to reduce eutrophication risk and improve the ecology and water quality in the mid 1980s, STW inputs needed to be targeted.

The SRP concentration/flow relationship for the period 1996 to 2000 shows that the introduction of limited improvements at five of the largest STWs in the late 1980s and early 1990s resulted in an improvement in water quality, with a reduction in the number of observations greater than 200 μg SRP l^{-1} from 28 in 1981 to 1985 to 15 in 1996 to 2000 (Fig. 6b). The Q_e value was reduced to 7.22 $m^3 s^{-1}$, indicating that the river was less point source dominated than in the 1980s. The model estimated that annual SRP inputs to the river from point and diffuse sources were approximately equal, contributing 49 and 51% respectively. Using traditional source apportionment methodologies that only provide annual load estimates, it would seem logical that either point or diffuse sources could be targeted at this site to improve water quality. However, the Load Apportionment approach identifies that the majority of high SRP concentrations and 86% of the summer growing period data points (between April 1st and September 30th) still occurred at flows less than the Q_e value, when point source inputs were dominant (Fig. 6b). Therefore, it would be much more effective to target STW inputs to reduce eutrophication risk.

The effectiveness of reducing STW outputs in the period 2001 – 2005 is demonstrated in Fig. 6c. The maximum SRP concentrations observed at the lowest flows reduced from 300 to 200 μg l^{-1}, and the SRP concentration was regularly below 90 μg l^{-1}, which is the P limiting concentration for algal growth in the River Frome (Bowes et al., 2007). This suggests that algal growth would have decreased following the STW improvements in 2002, due to P limitation for periods during the growing season, resulting in an improved ecology and a reduced risk of eutrophication. The Q_e value was reduced to 3.66 $m^3 s^{-1}$, and the proportion of the total annual SRP load derived from point source inputs was reduced to 33% (Table 3). The number of days when the mean daily river flow was below the Q_e value during the April 1st to September 30th plant growing season was reduced from 86% in 1996 – 2000 to 42% in 2001 – 2006, indicating the step reduction in point source input. This would imply that diffuse input reductions should be targeted in the future, as they make up two thirds of the annual SRP load. Despite this, Fig. 6c shows that the majority of high concentration data points during the summer growing season are still point source dominated, as most occur at flows lower than the Q_e of 3.66 $m^3 s^{-1}$. The high SRP concentrations that occur at flows $>Q_e$ are more probably associated with random pollution incidents (Jordan et al., 2007), as they occur at medium, rather than high flows. They could also be due to large hysteresis patterns during small storm events following a dry period (Bowes et al., 2005b; Stutter et al., 2008), as many of these outlying points occur in October, during the first storm event after the summer low flow period. Therefore, phosphorus mitigation in the River Frome in the future should still be focussed on STW reductions, and, to a lesser extent, identifying and controlling the non-flow-related pollution incidents. Diffuse pollution controls should be aimed at inputs occurring during storm events following a prolonged dry period in which P has had time to accumulate on the floodplain.

3.6. *Application of the Load Apportionment Model to other geographical regions*

This modelling approach has already been successfully applied to a wide range of rivers of varying size, land use and population density across England (Bowes et al., 2008), and this present study extends the application of this model to identifying changes in phosphorus source apportionment over a timescale of decades. However, care must be taken when applying this methodology to other regions of the world, with different land uses, industry and climate. For instance, in England with its high human population density, constant inputs of phosphorus to rivers will be dominated by STW effluents and industrial inputs. In areas with lower population density, the majority of constant non-rainfall-related P inputs could be due to agricultural practises that are not common in England, such as intensive fish farming and crop irrigation. It is therefore important to identify agricultural land uses that could contribute to these constant inputs, prior to applying the Load Apportionment Model. This potential weakness could actually be a strength, as it could provide a rapid and simple method to quantify such inputs in catchments of low population density.

4. Conclusions

The 40 year SRP concentration and flow record for the River Frome is a rare data set which allows the impact of past changes in catchment characteristics to be inferred. Although catchment information prior to 1990 is sparse (which often limits the usefulness of the SRP data), the application of the Load Apportionment Model greatly increases the value of such a data set, as it can determine the relative contributions of point and diffuse source inputs to the total river load, without the need for catchment data. This approach has shown that the changing SRP concentrations and the inferred contributions from point and diffuse inputs appear to agree with known changes in catchment population, land use and sewage treatment. This study has highlighted the importance of STW inputs, even for a largely rural catchment such as the River Frome, and shown how the introduction of phosphorus stripping at the larger STWs under the European Unions Urban Waste Water Treatment Directive (EEC, 1991) can result in a major improvement in water quality, both in terms of annual SRP load reduction, but also reductions in SRP concentration during the ecologically sensitive plant and algae growing period.

The Load Apportionment methodology has been shown to be a valuable tool in predicting changes in SRP concentration through the annual cycle that would result from the implementation of a P reduction measure. This ability to predict

P concentrations through the summer growing period, in conjunction with knowledge-based, maximum SRP targets obtained through algal limitation experiments (Bowes et al., 2007), will give catchment managers the ability to determine how much reduction of the SRP concentration is required, and what remediation strategies would best deliver this, to facilitate a reduction in eutrophication risk.

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